# AI.310

## Robot Localization and M

### COURSE OVERVIEW

Knowing where you are and what is around you: localization, mapping, and SLAM.



ILLUSTRATIVE HERO IMAGERY

ROBOT LOCALIZATION AND MAPPING

1

SESSION

 60 MINUTES

## ORIENTATION: WHERE AM I?

**KEY TOPICS**

- Localization vs. mapping
- The localization problem
- Map representations
- SLAM overview

**TEACHING FOCUS**

Ground the robot in its environment.

▲ **KEY TAKEAWAY** Navigation starts with knowing where you are.

ROBOT LOCALIZATION AND MAPPING

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SESSION

60 MINUTES

## LOCALIZATION

**KEY TOPICS**

- Monte Carlo localization
- Particle filters
- EKF localization
- Global vs. local localization

**TEACHING FOCUS**

Estimate robot pose from sensors.

▲ **KEY TAKEAWAY** Filters track pose over time.

ROBOT LOCALIZATION AND MAPPING

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SESSION

60 MINUTES

## MAPPING

**KEY TOPICS**

- Occupancy grid maps
- Feature maps
- Topological maps
- Map building

**TEACHING FOCUS**

Represent the world for a robot.

**KEY TAKEAWAY** Maps are the robot's memory of space.

ROBOT LOCALIZATION AND MAPPING

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SESSION

60 MINUTES

## SLAM

**KEY TOPICS**

- Simultaneous localization and mapping
- EKF SLAM
- FastSLAM
- Graph-based SLAM

**TEACHING FOCUS**

Build a map while locating in it.

**▲ KEY TAKEAWAY** SLAM solves the chicken-and-egg problem.

5

SESSION

60 MINUTES

# LOOP CLOSURE & RELOCALIZATION

**KEY TOPICS**

- Detecting revisited places
- Loop closure
- Kidnapped robot problem
- Global localization

**TEACHING FOCUS**

Correct drift and recover from loss.

**KEY TAKEAWAY** Loop closure keeps maps consistent.

ROBOT LOCALIZATION AND MAPPING

6

SESSION

60 MINUTES

## LOCALIZATION & MAPPING & REVIEW

**KEY TOPICS**

- SLAM demo
- Course recap and assessment
- Toward robotics planning (AI.311)
- Mapping exercise

**TEACHING FOCUS**

Run SLAM on a mobile robot.

**KEY TAKEAWAY** A mapped world is a navigable world.